

Early Detection of Epidemics with mechanistic disease information and human behavior – A Machine Learning approach featuring Risk Index, SIR and Government policies.

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I. Abstract— The early detection of epidemics is of importance to arrange the necessary public health interventions in the form of measurements. This research paper uses a proposed machine learning framework that uses Risk Index (RI) for the purpose of classifying transmission trends in three different categorical labels: Decrease (L0), Maintain (L1), and Increase (L2). This original approach managed to achieve an outstanding accuracy (over 94% accuracy) but had some gaps such as the lack in transmission dynamics information and government issued policies, both are necessary elements that are needed to model how disease spread and how humans respond to diseases respectively. The following proposed approach introduces features extracted from the Susceptible-Infected-Recovered (SIR) model which include transmission rate, recovery rate, basic reproduction number, effective reproduction number, growth rate, and doubling time. And incorporates 21 policies from the Oxford COVID-19 Government Response Tracker which include stringency index, vaccination policy and school closures. Adapting to the same 35 day sliding window from the previous framework (21 days calibration, 14 days prediction) and the same Machine learning classifiers (XGBoost, Random Forest, Support Vector Machine), for the mean across models our proposed enhanced model achieves a classification accuracy of 94.52% for the combined approach of RI + Policy feature set, slightly outperforming the All Features set model (94.17%) and the RI only feature set (93.35%). Additionally, the inclusion of SIR and extra policy features allow this framework to provide interpretable insight on epidemic outbreaks. Feature importance analysis shows that the top 3 features are: mean normalized case count, debt/contract relief, and restrictions on gatherings. Model also achieves an early-warning AUC of 0.98 allowing this framework to surpass the original one in terms of practical utility. The results gathered from this paper suggests that combining statistical trend indicators such as RI with epidemiological dynamics methods such as SIR and

human behavior dynamics in the form of policies yield a robust, interpretable and actionable outbreak detection algorithm that can be used accordingly by public health and authorities to minimize the damages from epidemic spreading.

Keywords— Outbreaks, COVID-19, Epidemic, Epidemiology, SIR, Risk Index, Machine Learning, Public health, Policies, Early Warnings, Vaccination, Government measures.

II. INTRODUCTION

The coronavirus disease (COVID-19) pandemic, originated by the SARS-CoV-2 virus has caused one of the most impactful and disruptive global health crises on the 21st century. Many confirmed cases worldwide leading to the death of millions of people and the disruption of many countries' economies and daily activities. Due to emergence of different transmissible variants such as Delta and Omicron many health care institutions were working under very non ideal circumstances, and governments needed to implement measurements to slow the spread of the virus while minimizing losses to their economies. In this context, we can attribute meaningful value to a tool capable of predicting the spread of the outbreak to make timely and actionable decisions. Research has shown the use of Machine Learning as a powerful tool for forecasting, and this is no exception for epidemic spreads. Cho et al. Proposed a ML based framework that uses Risk Index (RI) to quantify the change in transmission trends between a calibration (21 days) and a prediction period (14 days) totaling 35 days per window, across 953 rolling windows. The resulting RI value is later used to create labels to decrease (L0), maintain (L1), or increase (L2) dividing the RI values into terciles. The original approach then utilizes ML (XGBoost, Random Forest, SVM) with eight main features (mean of the calibration period, slope, standard deviation, proportions of Delta and Omicron variants and policy levels) to predict these labels from RI. While this proposed method achieved very high accuracy (over

94% for XGBoost and RF) it does not address the need to identify essential epidemiological dynamics such as the ones contained in the Susceptible-Infected-Recovered (SIR) and the impact of government policies such as vaccination, school closures and mobility restrictions which were missing in this original framework. So, despite the high accuracy that this research has, it lacks mechanistic information since RI is a purely statistical method and cannot capture epidemiological essential parameters that describe how diseases work. While the authors included a single policy on social distancing level, they neglected many other informative intervention policies that are available in sources such as the Oxford COVID-19 Government Response Tracker (OxCGRT). The original work was more focused on outbreak detection rather than early warning evaluation, which meant they could not quantify lead time or assess the transferability aspect across the phases of an epidemic. Additionally, the paper omitted the intervention of Vaccination despite them having a critical task in controlling the pandemic.

For this paper we have extended the framework by incorporating two main novelty features:

SIR features: For each of the 21-day calibration windows across the 953 rolling windows, we estimate the parameters of a regular SIR model by using a reverse engineered SIR approach. These features are transmission rate, recovery rate, basic reproduction number, effective reproduction number, growth rate, and doubling time.

Government policy features: We extracted from the OxCGRT dataset 21 policy variables which include stringency index, school closures, vaccination policy, mobility restrictions. We then applied a 14-day delay to account for the lag between implementation and effect. We aggregated these features over the same 35-day windows and got a total of 84 features as each policy had mean, slope, start and end.

III. Literature Review

The main literature review approach was analyzing three research papers from various sources. All of them were written by researchers or experts in the field and were peer reviewed before being put into their respective research journals. Universal to all three papers were machine learning methods such as Logistic Regression, SVM, Random Forest, and others, which makes sense for predictive modeling and studying epidemiological trends. Datasets that were used were also preprocessed and had exploratory data analysis run on them.

The first paper was from the NIH, written by Shan Gao, Amit K Chakraborty, Russell Greiner, Mark A Lewis, and Hao Wang. It was the *Early detection of disease outbreaks and non-outbreaks using incidence data: A framework using feature-based time series classification and machine learning*. The paper uses incidence time-series data to develop a machine-learning framework for the early detection of both disease outbreaks and non-outbreaks. The raw incidence data, which in this instance, is the counts of infected individuals over time. The data is turned into statistical and early-warning signal features (EWS) to then be used to train supervised classifiers to distinguish between outbreak and known non-outbreaks. This paper uses Logistic Regression, Support Vector Machines (SVM), K-nearest neighbors, as well as Gradient-Boosting machine (GBM). These methods ran on synthetic data from what is called an SIR model (Susceptible, Infected, Recovered). That model's calculation uses S , I , and R , as well as time t , recruitment rate of susceptible population Λ , and the transmission rate at time t is $\beta(t)$, μ is the death rate, and α is the recovery rate. The issue the authors tried to address is that most existing methods focus only on outbreak detection while ignoring non-outbreak prediction. Furthermore, traditional EWS methods are sensitive to noise and are often unreliable. The results showed that outbreaks and non-outbreaks produce statistically distinguishable signatures prior to outbreak onset, and this allowed for accurate early classification. Overall, the framework provides a strong-performing possible alternative to traditional epidemiological modeling and a generalized framework. It enables the identification of non-outbreaks, helping to reduce unnecessary outbreak alerts. The author taps into previous work on early warning signs (EWS) literature. Past studies showed that systems approaching tipping points show rising autocorrelation, changing values of skewness, and escalating kurtosis. Additionally, the types of past work this paper analyzes are ML approaches that are often disease-specific and focused on prediction after outbreak onset. The past work sheds light on the problem by highlighting the need for a unified, general framework. The Time Series Classification that this paper focused on is applied systematically to disease outbreaks. This paper is important to discipline, especially epidemiology, since it offers research and practical early warning tools for public health. Since focus on this space is on reactive outbreak detection, this paper shifts it to proactive risk assessment. Limitations include how the synthetic data used in this may not be as informative and complex as the data that could have been gathered. Furthermore, this model only tested COVID-19 (SARS) which might have unique properties among diseases. Lastly, the predictions are limited to only binary classifications,

which is predicting if it will be an outbreak or non-outbreak.

$$\begin{aligned}\frac{dS(t)}{dt} &= \Lambda - \beta(t)S(t)I(t) - \mu S(t), \\ \frac{dI(t)}{dt} &= \beta(t)S(t)I(t) - \alpha I(t) - \mu I(t), \\ \frac{dR(t)}{dt} &= \alpha I(t) - \mu R(t),\end{aligned}$$

Paper 2

Next, was our parent paper, which is the *Detection of COVID-19 epidemic outbreak using machine learning* written by Giphil Cho, Jeong Rye Park, Yongin Choi, Hyeonjeong Ahn, and Hyojung Lee, and published in Frontiers. Similar in subject to the previous paper, this paper used epidemiological data as well, but instead to develop a Machine learning method of estimating COVID-19 transmission trends and detecting the start time of new outbreaks. It was undertaken because existing predictive tools had limited capability in early detection. It addressed such difficulty of using traditional statistical methods, which do not adapt to factors like policy changes or new variants. Therefore, they wanted to establish a revised ML method that ultimately provides more accurate tools that could improve outbreak preparedness and response. The methods used here are a combination of Support Vector Machine (SVM), Random Forest (RF), and XGBoost (XGB). These were combined with the risk index concept, which is a measure of tracking changes in transmission patterns. The findings include that RF provided the best outbreak detection performance, followed closely by XGB. The Active(L2) achieved a high accuracy of more than 94% in classifying transmission trends. This paper is important to the discipline and contributes to epidemiological studies by introducing an ML framework that does in fact detect outbreak signals earlier and with high accuracy. This is helpful for future pandemics. The authors built upon existing ML methods and used epidemiological data from Korea. It referenced prior work that used SVR/LSTM, ARIMA, and decision trees to model COVID-19. The study also transformed time-series patterns into features for classification. Limitations in this study include how the data only spans two years, or how the study did not consider the effects of vaccination. Furthermore, the data was only from Korea, so it is not generalized.

RF

	L0	L1	L2
L0	93	2	0
L1	4	90	1
L2	0	5	91
	L0	L1	L2

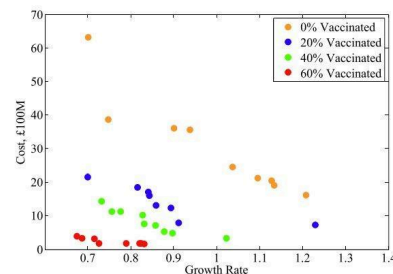
Predicted label

Figure 2 is the confusion matrix showing RF with the best results, which was consistent with our findings.

Paper 3

The third paper was titled *Modelling the effect of vaccination on the spread of COVID-19 via a novel evolutionary ensemble learning algorithm*, written by Mohammad Hassan Tayaran Najaran and published to the Science Direct Journal. The paper used vaccination rates data and government policy data to develop an ML method of predicting COVID-19 case

growth. The reason why this work was necessary was because when handling and intervening in the COVID-19 pandemic, vaccine rates affected the usefulness of policies. The traditional predictive models usually do not consider or even fail to effectively combine vaccination data with policy impact. They used two algorithms. One is a novel ensemble learning algorithm to generate diverse base learners, meaning multiple different models so that their combined predictions are more powerful. The other is an evolutionary optimization algorithm that uses a predictive model as a fitness function to find policy sets that minimize cost to society and infection rates. There were 20+ policies that ranged from relating to social distancing and contact tracing to public transport and vaccination. Overall, the proposed ensemble model outperformed existing methods in prediction accuracy, and there was an improved identification of effective policy sets based on different vaccination levels. Furthermore, they found that combining ML with optimization techniques can actually guide decision making, either at the state or town level. This paper applies machine learning to practical public health applications by introducing a single framework for both vaccination data and policy so that policy makers can make decisions that consider pandemic control and impact cost. It used the theoretical perceptive and concepts of ensemble learning by combining multiples models to increase prediction performance. Past models often used single predictors, and commonly used ensemble methods right now improve the performance but lack the evolutionary diversity and multi-objective optimization. This work addresses shortcomings of prior models that don't combine policy optimization and vaccination influence. Limitations include how it had a limited ability to capture real-world human behavioral factors that influence transmission, and how the findings based on past data may not generalize well to future variants or new pandemic conditions since the model is trained on historical pandemic data.



IV.

Novelty and Contributions:

The original paper used Machine Learning techniques (which can be categorized as Support Vector Machine, Random Forest, and XGBoost) and just the risk index on top of that for their model. Additionally, they added a single policy for social distancing. All of this was done in their training.py file. Our contribution was combining that with the SIR Model (Susceptible-Infected-Recovered) plus the different social and government policies outlined in the third paper. This was done in the file we created, enhanced_training.py, as well as four other supporting files that work together to make the new model run smoothly since the data and the pipeline changes too. As for metrics, we still focused on mainly the accuracy, followed by precision, F1-score, recall, and support. On the new model, The Test Accuracy Mean across all models had an accuracy of 94.17% which was the All Features one. The highest was RI + Policy at 94.52%.

New methods breakdown:

SIR Feature extraction: For the 953 rolling windows, we extracted the daily case counts from the 14-day prediction period. By using reverse engineering approach, we were able to invert the classic SIR model in a way such that we could input the daily case counts and obtain the epidemiological parameters as output. To do this we implemented a constrained optimization algorithm that can ensure epidemiological plausibility through minimizing weighted mean squared error between cases and the model predictions, and by imposing bounds on some of the parameters. The computed parameters were the following:

- Transmission rate (beta): Daily infection rate per infected individual
- Recovery rate (gamma): Daily recovery rate (1 / infectious period)
- Basic reproduction number (beta / gamma): Expected secondary infections in completely susceptible populations
- Effective reproduction number: Expected secondary infections when accounting for immunity
- Epidemic growth rate (beta – gamma): Net daily growth of the infected population
- Doubling time: Required days needed for cases to double

Policy Feature Extraction: We downloaded the OxCGRt dataset for South Korea which possesses records on 21 policy indicators for the daily basis (stringency index, school closures, testing policy, vaccination policy, mobility restrictions). The main challenge was to align the 35-day window structure with the policies. To address this, we applied a 14-day delay, meaning we shifted all policy variables forward

by 14 days to account for their influence in disease spread. Then for each policy we computed the mean, the slope, the start, and the end value after the delayed window. This step left us with 84 policy features. However, six mobility related policies only contained zero values and were removed (24 features). For the remaining 60 features, 3 additional features were excluded in the preprocessing step due to zero variance or high collinearity. So, the final model contained 57 policy features.

Feature set accuracy comparison:

Considering the fact that we had features from many different sources, we used the ML models to test the accuracy of 7 different feature set configurations. One of these configurations was the “All Features set” which included the following breakdown of features:

- RI features: 8
- SIR features: 6
- Policy features: 57
- Total: 71

V. Implementation Strategy

After doing our research on the reference paper 1 and replicated the baseline code for our parent paper (paper 2) we implemented a script called SIR_Feature_Extraction.py in which we processed the Korean case data in order to fit in an Inverse SIR model that would extract all the parameter as output from the daily cases as input. 6 features were obtained in this process. Then after researching paper 3 we decided to create a script called Policy_extraction.py in which we preprocessed the data from the Oxford dataset so it would fit on our 953 rolling windows, we converted the policies that were nonzero into 4 features for each variable and we end up getting 57 features. In this same script, we combined all features from SIR, RI, and the ones obtained from the policies to create the whole set of 71 features. This enhanced_with_policies file was then used to create the enhanced_training.py script which we replicated almost identically but we used the new dataset with the combined 71 features instead. Then we developed two scripts, evaluate_all.py and Visualize_results.py which were implemented to reuse the models to do an analysis for different metric evaluation, evaluate_all.py focused on generating these results evaluation and Visualize_results focused on generating figures for us to use in our report

VI. Methodology:

How the Machine/Deep Learning works:

For the Support Vector Machine, a Radial Basis Function kernel SVM was used to handle non-linear separability between outbreak classes. The RBF kernel maps the input features into a higher-dimensional space, where the model can make a maximum-margin hyperplane that separates the classes. The SVM

identifies a subset of training points, which are the support vectors, which define the position of the decision boundary. Then it makes the hyperplane that maximizes the margin between classes, leading to improved generalization on unseen data. The key hyperparameters were $C = 50$, $\gamma = 0.3$, and Kernel = RBF for the kernel. C and γ control the trade-off between margin width and classification error, as well as the influence of individual training points.

Next, Random Forest is an ensemble learning method that creates multiple decision trees using bootstrap sampling and feature randomness. The final predictions are made through majority voting across all the trees. RF is good for handling nonlinear interaction, robustness to noisy features, and for feature importance estimation, which tells how much each feature contributes to reducing classification error. Here, the key parameters are, number of trees = 85 and the maximum depth = 14.

The third ML method was XGBoost, and this is a gradient boosting method (extreme) that builds trees sequentially, and each new tree corrects errors from previous ones. It is optimized for speed and performance, as it focuses on minimizing a specified loss function using gradient descent. At each of the iterations, the model calculates the gradient of the loss function with respect to the current predictions and fits a new tree to approximate this gradient. This iterative process allows XGBoost to capture subtle patterns and interactions in the data that may not be detected by other models. Also, XGBoost incorporates regularization techniques, such as tree depth constraints and shrinkage (learning rate), to prevent overfitting. XGBoost is strong for capturing complex feature interactions, handling data that is structured tabularly, and also for achieving high predictive accuracy in small-to-medium datasets. The key parameters here are $\text{max_depth} = 7$ for 7 levels of splits, and $\text{n_estimators} = 110$ for the 110 trees or boosting rounds.

For each of the trained models, the class probability estimates can be calculated following: $P(y=k|x)$, where x is the feature vector for a given window and k (0,1,2) corresponds to the outbreak levels. These probabilities give a measure of the model's confidence in assigning a sample to each class. You get the final prediction by selecting the class with the highest probability (argmax rule). Beyond classification, these probability outputs are important for evaluating model performance using ROC-AUC and for enabling early warning analysis, as rising probabilities for the "Increase" class can indicate an impending outbreak even before a class switch occurs.

VII. Results:

Model Performance:

The performance tests done on our different feature sets indicate that RI + Policy performed the best with an accuracy of 0.9452. Standing slightly below, we have the All Features set which achieved an accuracy of 0.9417. This difference is almost negligible, meaning the addition of SIR held similar results while still contributing with valuable variables for interpretation on outbreaks. We can notice how the difference between SIR and SIR + Policy is quite significant, demonstrating that SIR parameters are not strong predictors on its own, but work well combined with the policy contents.

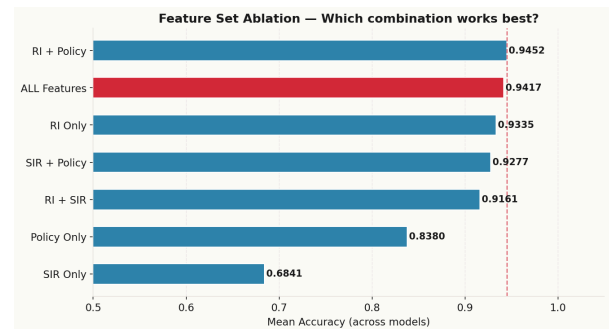


Figure 4: Feature Set Test Accuracy Mean Comparison across models chart.

Additionally, the following table provides the best test for model performance on a subset of 286 windows, showing how RF was the top performer and contributed significantly to raising the mean value across the models in different metrics, noticeably, test accuracy at 0.9720 and F1-L2 (outbreak detection) at 0.9856 which tells us that the RF model is almost certain when there is an increasing outbreak. F1-L1 values were slightly lower compared to F1-L2 but this is somewhat expected as the transition between L0 (declines) and L2 (rises) is known to be ambiguous.

Table 1 – Model Performance Summary (Test Set: 286 windows)

Model	Test Acc	CV Mean	CV Std	F1-L0	F1-L1	F1-L2	Macro F1	Weighted F1
SVM	0.9161	0.8950	0.0230	0.9153	0.8770	0.9519	0.9147	0.9168
Random Forest	0.9720	0.9370	0.0176	0.9721	0.9565	0.9856	0.9714	0.9722
XGBoost	0.9371	0.9296	0.0100	0.9667	0.8977	0.9444	0.9363	0.9367

Figure #5: Model Performance table across different metrics done on a 286 rolling windows subset

Feature Importance:

Policy variables are dominating the aggregated importance chart by a lot, with them including the top 2 variables. Followed by the RI variables and then followed by the SIR variables. Considering that there were 57 policies, 8 RI and 6 SIR features, we can at least see that policy variables did not completely overshadow RI and SIR. RI gives very informative signals Δ^c with the current variant profile, μ^c

helps the model realize how further into the outbreak it is, and β^c captures the slope of this trajectory. Although Simple these are powerful tools for the models to read. Then SIR growth_rate_max the top performer for SIR can tell the model the highest point for rate transmission which allows a better understanding of how fast and aggressive the spread is going to be. Finally, the top 2 policy variables E2 Debt/contract relief, and C4M gathering restrictions. The E2 Debt/contract relief policy acts as an indicator for the model when debts get relieved at times of great restriction, so the model interprets that when debt relief is active a really strong phase of the outbreak is approaching, similarly when these are not active it knows a recover period is happening as debt relief changes. Gathering restrictions were expected to be relevant as the model learns how restrictions get stronger as waves are starting, and they get relaxed as they enter a recovery phase from an outbreak. We could expect stringency index to be higher since it measures the strictness of all policies, but since these are already being analyzed, the whole policy set does not give as much signal as compared if we were not analyzing individual policies. Vaccination has importance but the reason is not higher as we first guessed is cause large outbreaks were occurring when vaccination was also at its highest, the model never got to learn the high spread vs recession spread pattern as the vaccination in Korea was continuous throughout the rolling windows.

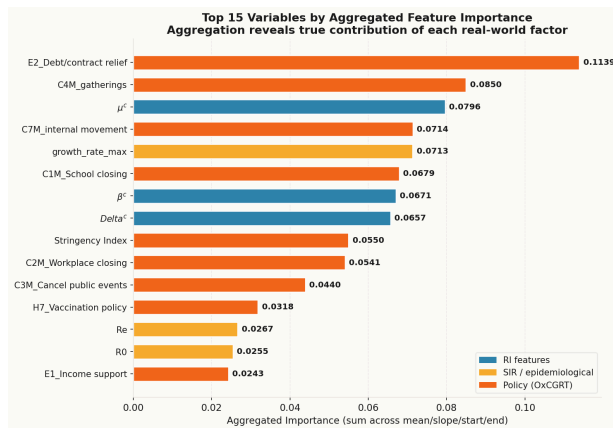


Figure #6: Feature importance chart for Top 15 Variable aggregated by feature importance

Outbreak Detection:

The predicted L2 label was consistent with the ones reported in the paper. Detecting a total of 7 outbreaks across the 953 rolling windows

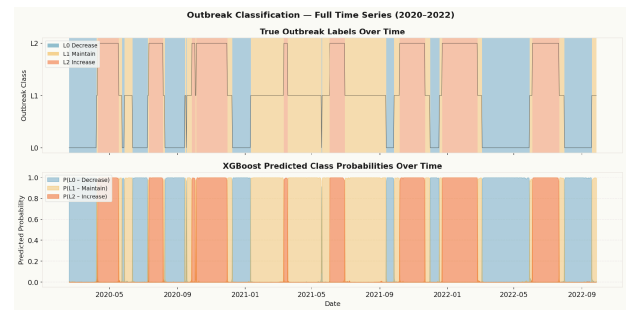


Figure #7: Outbreak Classification Full Time Series graph

Early Warning:

The early warning AUC of 0.9795 is the primary value obtained from this metric. For the lead time aspect we can see from the graph that the dashed line along the graph has a 0.3 threshold value, from June to July 2022 transition we first get an alert early June on the confirmed L2 label happening after mid-June that continues until early August. This can be considered a lead time of 2 weeks of advance for an outbreak period.

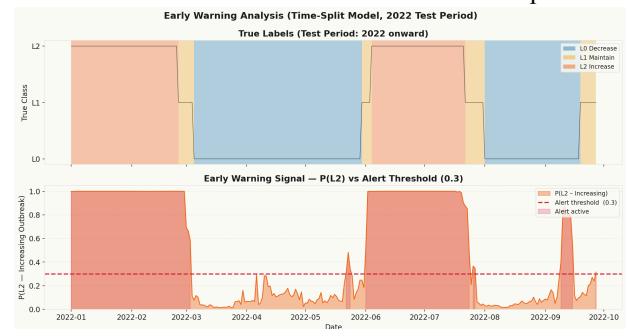


Figure #8: Early Warning Analysis, True labels and Early warning signals comparison graph

Transferability:

From this section, we can see how the difference between a subset of the original time split and a random split is only lower by 0.1084. So, in the worst-case scenario, the model would be able to classify with a 0.8333 test accuracy. Since the time split is based on the later phases of covid, we can somewhat attribute this drop not cause of model error but because of temporal shift, due to the change in covid variant to omicron and its dynamics.

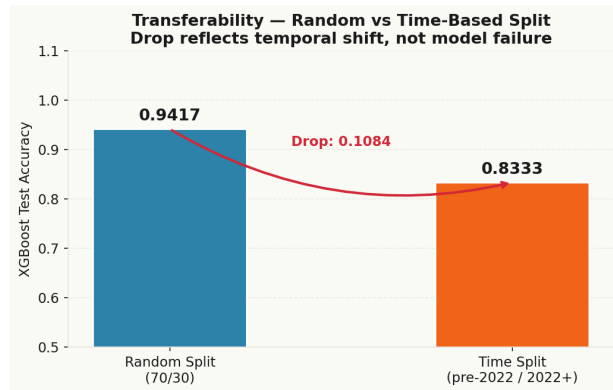


Figure #9: Transferability chart, Random split vs Time based split

VIII. Pros and cons

The main advantage of our approach is combining the Risk Index, SIR, and Policy extraction to create a more powerful and robust predictive model that takes much more into account than the model in our parent paper. There are now multiple perspectives on disease dynamics to capture a bigger picture of how the outbreak will play out. For scientists, public health officials, and policymakers, this will give more interpretability and actionable insights.

Another advantage was still retaining high performance across the models. After the machine learning process was applied, we still obtained an accuracy of more than 94 percent on the important metrics.

Additionally, using three models in the Machine Learning Process (SVM, XGBoost, and Random Forest) helped with robustness of the model. Just using one would not have been able to effectively cover nonlinear relationships, interactions between variables, and noisy data well.

Also, the inclusion of feature importance analysis provides transparency by identifying which variables are the most influential in the prediction, and then those high importance variables can be evaluated for making public health policy and taking action when an outbreak does arise.

One con is that our approach introduced increased complexity. Adding an SIR and numerous policy features significantly expands the feature space since there are now 70+ features to consider. Without carefully managing this, it may lead to overfitting,

Also, even though using cross-validation mitigated the risks of overfitting, the model still became computationally complex and harder to compare relative to the framework with just the Risk Index.

Additionally, we only know so much about the Oxford Dataset (OxCGRT), and although we did our own verification on the accuracy of that data, we could still be unsure about the exact specifics of how it was collected or if it is truly unbiased. How well the data translates into the results we obtain depends on South Korea's covid cases, which we then use to make generalizations, which again highlights the same data dependency concerns.

Lastly, estimating SIR parameters introduces a potential source of error. Since we reverse engineered it, model assumptions can propagate into the features. Also, the classical SIR model does not account for factors such as asymptomatic transmission, heterogeneous populations, or spatial effects.

IX. Conclusion

In conclusion, we were able to successfully hybridize the findings and pipelines in previous work in our own research of Covid and epidemic spread. Our innovation was an even more powerful model to predict outbreaks of COVID-like airborne diseases. Though we went into the project not knowing whether adding features would ruin the performance and accuracy, we were ultimately able to obtain a comparable accuracy and result of metrics, showing how we were able to create a strong and more detailed model. Our findings can create actionable insights to public health policymakers, responders to public health emergencies, disease mitigation experts and agencies, or even everyday people interested in this research and the raw numbers from how well the model classified outbreaks or simply wanting to be prepared for the next pandemic. It is insightful to see how a model with so many different features worked to produce results through machine learning.

X. Future Work:

- Regression tools could replace or complement the three classifications (L0, L1, L2) with regression models such as XGBoost regressor and Random Forest regressor. This implementation could predict the exact number of new cases from the 14 days prediction period, with actionable insight on magnitude of outbreaks.
- Ensemble learning with a variety of Machine Learning models can be implemented to complement our current research with additions that add model diversity such as Logistic Regression, K Nearest Neighbor and Multi Layered Perceptron. Allowing to capture

different patterns.

- Inverse SIR experiments can be performed again like how we did for Korean data cases, and be used along local factors such as population density, local policies and other factors to predict daily cases from a different setting than Korea. Comparing this Synthetic Data with real reported data would allow us to study the importance of local factors.

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